Lecture 2: Samples, standard errors and confidence intervals

Empirical Methods for Law and Economics | EDLE

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Today's question

If we had observed a different set of cases,
how different would our estimate be?

- By the end of today you will be able to:
  - distinguish a population parameter from a sample statistic;
  - explain the difference between a standard deviation and a standard error;
  - interpret a 95% confidence interval correctly, and state the assumptions behind it;
  - recognise when observations are not independent, as in "A Fine Is a Price".

DISCUSS Retrieval

1. Why is a court automation score of 0 different from a missing value?

2. The mean enforcement time was 650 days, the median 575. What does that tell you?

3. What did the SD of 304 days describe?

ANSWERS Retrieval

- A score of 0 is an observed value (no automation features); a missing value means the score is unknown.

- Mean 650 above median 575: a long right tail of slow economies pulls the mean up.

- The SD of 304 days described how different the 191 economies are from one another, in days. It said nothing about uncertainty.

From describing to estimating

- Thursday: we summarized the indicators for the 191 economies in our file. Our mean described that file; it was not an estimate from a random sample of court cases.
- Typical legal research: a court registry holds 10,000 closed commercial cases; we can read and code only 25 or 100 files.
- We want the average duration of *all* 10,000 cases. We only see a sample.
- Two different problems:
  - **sampling uncertainty**: another random sample would give a different mean;
  - **selection**: the files we can read may differ systematically from the others (e.g. only cases that reached judgment).
- Statistics quantifies the first. A larger sample does not cure the second.

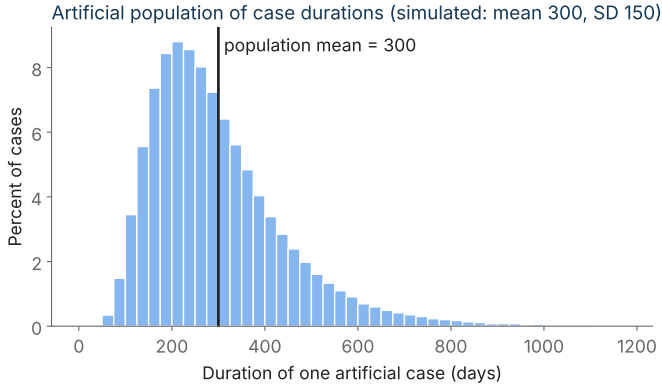
Parameter, statistic, estimator

	Population (fixed, unknown)	Sample (varies)
Mean	μ	$\bar{y}$
Standard deviation	σ	s
Proportion	p	$\hat{p}$

- An **estimator** is a rule ("average the sampled durations"); an **estimate** is the number it gives in one sample.

- Same rule, different sample $\Rightarrow$ different estimate.

PREDICT An artificial population where we know the truth



- Simulated case durations: right-skewed like Thursday's data; mean $\mu = 300$ days, SD $\sigma = 150$ days.
- We specify the population mean as 300 days. In real research, the population mean is usually unknown: that is what we want to estimate.

Draw 25 cases from this population: how far from 300 will their mean typically be?

PREDICT Standard deviation versus standard error

With larger samples, does the SD of the cases fall? Does the SE of the mean?

ANSWERS Standard deviation versus standard error

- **SD** (σ, estimated by s): how much individual cases differ from one another.
- **SE**: how much the *sample mean* would differ across repeated samples.

$$SE(\bar{y}) = \frac{\sigma}{\sqrt{n}}, \qquad \text{estimated by } \frac{s}{\sqrt{n}}.$$

- With $\sigma = 150$: $n = 25 \Rightarrow 150/5 = 30$ days; $n = 100 \Rightarrow 150/10 = 15$ days.
- Quadrupling n halves the SE. The cases themselves do not become more similar: σ stays 150.

Why $\sqrt{n}$? (and why independence matters)

- If $y_1, \ldots, y_n$ are independent with variance σ^2:

$$\text{Var}\left(\sum_i y_i\right) = n\sigma^2 \quad \Rightarrow \quad \text{Var}(\bar{y}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \quad \Rightarrow \quad SE = \frac{\sigma}{\sqrt{n}}.$$

- The first step uses independence: positive and negative deviations of different cases partly cancel.

- If observations are positively correlated (for example, cases sharing a judge), they provide less independent information. The simple formula can then understate uncertainty.

STATA Guided Stata: the supplied simulation

```
do "do/00_setup.do"
do "do/ready_to_run/02_sampling_demo.do"
use "results/tables/02_sampling_results.dta", clear
list n rep samplemean sd se if n==25 & rep<=5, noobs
```

- The program draws 2,000 samples each of size $n = 5$, 25 and 100 and records, for each sample, its mean, SD, SE and 95% interval.
- One row of the results file = one simulated **sample**, not one case.
- Read the five displayed rows: all have 25 cases, but each is a different sample. Their means differ even though the population is unchanged.

STATA Reading Stata: two different kinds of variation

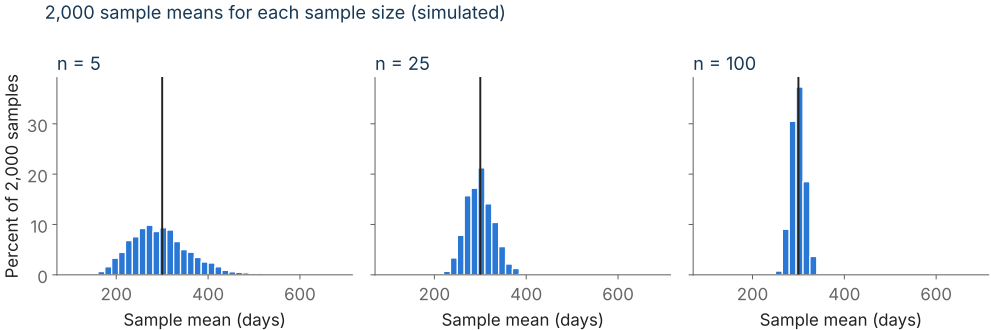
```
summarize samplemean if n==25
summarize se if n==25
```

- First command: restrict the results to samples with 25 cases. The displayed **Std. dev.** measures variation across the 2,000 sample means: approximately 30 days.
- Second command: the variable `se` stores the estimated SE from each sample. Read its **Mean**: this averages those estimated SEs.
- Repeat both commands with `n==100`. The means should vary less. Do not confuse the output column **Std. dev.** with the variable named `sd`.

PREDICT What the simulation shows

Predict the SD of the 2,000 sample means for $n = 5$, 25 and 100.

ANSWERS What the simulation shows



	$n = 5$	$n = 25$	$n = 100$
Theoretical SE, $150/\sqrt{n}$	67.1	30.0	15.0
SD of the 2,000 sample means (approx.)	67	30	15

The central limit theorem (informally)

- For independent draws with finite variance, the distribution of $\bar{y}$ becomes approximately normal as n grows, *whatever* the shape of individual outcomes.
- That is why we can build intervals with normal or t critical values.
- "Approximately" and "as n grows" matter: with $n = 5$ from a skewed population the approximation is still poor.
- With dependent observations the mean can still be approximately normal, but its SE is no longer $\sigma/\sqrt{n}$. Extremely heavy tails are a further problem.

A 95% confidence interval

$$\bar{y} \;\pm\; t_{n-1,\,0.975} \times \frac{s}{\sqrt{n}}$$

- Centre: the estimate. Width: the estimated SE times a critical value.
- The critical value is a multiplier that Stata supplies. It depends on $n-1$ degrees of freedom: 2.064 for $n = 25$, approaching 1.96 for large samples.
- Example: $n = 25$, $\bar{y} = 312$, $s = 150$: $SE = 30$, $t_{24} = 2.064$,

$$312 \pm 2.064 \times 30 = [250.1,\ 373.9] \text{ days.}$$

- With $n = 100$ and the same $\bar{y}$ and s: $SE = 15$, $t_{99} = 1.984$, interval $[282.2,\ 341.8]$.

DISCUSS What "95%" means

Which of these statements is correct?

1. 95% of cases fall inside the interval.

2. The true mean has a 95% probability of lying in this interval.

3. About 95% of intervals built this way, across repeated samples, would contain the true mean.

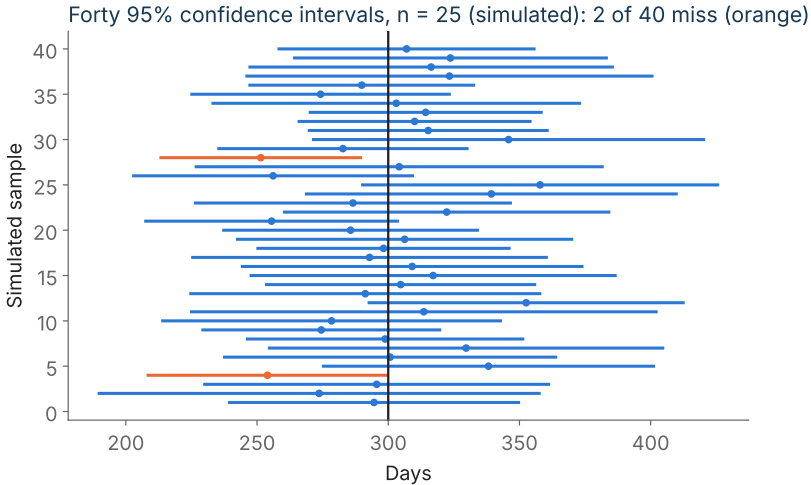
ANSWERS What "95%" means

- The true mean (300) is fixed. The *intervals* change from sample to sample.
- Under the required assumptions, about 95% of intervals across repeated samples would contain the true mean. For skewed outcomes this is an approximation that improves with sample size.
- It is a property of the **procedure**. A single realised interval either contains 300 or not; in real research we cannot tell which.
- It does *not* say that 95% of cases fall inside the interval, nor that the true mean has a 95% probability of lying in *this* interval.

PREDICT Forty intervals from forty samples

Out of forty 95% intervals, how many will miss the true mean?

ANSWERS Forty intervals from forty samples



Simulated: $n = 25$ per sample, true mean 300. Each interval uses a different sample. The fixed seed makes the supplied example reproducible.

Coverage depends on assumptions

	$n = 5$	$n = 25$	$n = 100$
Share of intervals containing 300 (approx.)	0.92	0.935	0.945

- The t interval has exact 95% coverage for independent normal observations. With our skewed population its coverage is approximate.
- With skewed outcomes and small samples the actual coverage falls short; it improves as n grows.
- There are two sources of discrepancy: an imperfect interval approximation and simulation noise. With 2,000 replications, small differences of around one percentage point need cautious interpretation.

DISCUSS What a confidence interval cannot repair

For each item, give an example from a legal dataset you know.

- Measurement: a precise estimate of a badly measured outcome.
- Selection: the sample does not represent the population of interest.
- Dependence: repeated measurements on the same units treated as independent.
- Substance: is a difference of 1 day or of 40 days legally or economically important? Report intervals in original units.

What a confidence interval cannot repair

- Measurement: Doing Business durations are expert estimates for one hypothetical case, not observed cases.
- Selection: a registry sample of cases that reached judgment misses those that settled.
- Dependence: many cases decided by the same judge; the same day-care centre observed week after week.
- Substance: a precisely estimated difference of one day that no litigant would notice.

DISCUSS Paper: Gneezy and Rustichini (2000), *A Fine Is a Price*

1. What does one row of Table 1 record, and what does one cell count? What does Figure 1 average?

2. Which centres receive the fine, how were they chosen, and in which weeks is it in force?

3. Why are the 200 centre-week counts not 200 independent observations of the effect of the fine?

4. Describe the pattern in Figure 1 in two sentences, without explaining *why* it happens.

A Fine Is a Price: the design

- Ten private day-care centres in Haifa, observed for 20 weeks. Outcome: number of parents arriving late to collect their child, per centre and week.
- At the start of week 5 a fine (NIS 10 per child for delays of at least 10 minutes) is introduced in 6 centres *selected randomly*; 4 centres serve as comparison.
- At the start of week 17 the fine is removed, without explanation.
- **Table 1**: one row per centre (with the number of children enrolled); one cell = late arrivals in that centre in one week.
- **Figure 1**: for each week, the average count across the centres of each group.
- Pattern: in fined centres late arrivals rise after the fine is introduced and stay high after it is removed; in the control centres no noticeable change.

DISCUSS From the $\sqrt{n}$ formula to the paper

- The fine was assigned to **centres**: 6 versus 4. The same centre is measured week after week.
- Weeks within a centre share parents, staff and routines: their outcomes may be correlated.
- Treating 200 centre-weeks as independent can overstate precision. There are 10 independently assigned centres; this does not mean we can simply substitute 10 into any SE formula.
- Constructive referee comment: "Please clarify how repeated weekly observations within centres are accounted for in the reported tests."

Which other objection would you raise as a referee? Internal or external validity?

ANSWERS Other referee objections

- **External validity**: ten private centres in one city, one level of fine (NIS 10). Would public centres, or a larger fine, behave alike?
- **Inference**: weekly counts within a centre are dependent; with 10 centres, tests should account for clustering.
- **Internal validity** is comparatively strong: fined and control centres, before and during the fine, with random selection of the fined centres. The design is a difference-in-differences.
- **Mechanism**: why does lateness rise? The authors read the fine as a price (Sections IV–V, next readings).

Laboratory

Student worksheet: `do/exercises/02_sampling_and_inference.do`.

1. Run the supplied simulation; say what one row of the results file is.
2. Core task: compare the SD of sample means for $n = 25$ and $n = 100$; then compare their average estimated SE. Explain both in days.
3. Compare the three histograms: which sample size gives the most concentrated means?
4. Compute the coverage for each n; explain why it is not exactly 0.95.
5. Extension for faster pairs: compare within-sample SDs; predict the SE for $n = 400$ and check it using the supplied commands.
6. Write two paragraphs: SD vs SE with your numbers; a correct 95% interpretation.

Exit check

1. SD or SE: which describes variation across individual cases?

2. If n is multiplied by four, what happens to the SE, and what to the SD?

3. In "95% confidence", what varies across repeated samples and what stays fixed?

4. In the day-care study, how many units were independently assigned to fine or no fine?

Next week

- Lecture 3: relationships between variables, hypothesis tests for means and proportions, association versus causation.

- Assessment: agree the paper or seminar for your referee report with me **by the end of next week**.